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Pinning device, casting unit as well as stretching unit

Abstract

A pinning device for the electrostatic pinning of a film on a chill roll is disclosed having a first coil unit, a second coil unit, a pinning region situated between the coil units, a high-voltage source as well as at least two electrodes. At least the first electrode of at least two electrodes is strip-shaped. The electrodes run from the first coil unit to the second coil unit in the pinning region and high voltage is applied to both electrodes by the high-voltage source.

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Background/Summary

CROSS RELATED APPLICATION

(1) This application claims priority to German Application DE 10 2022 118 971.6, filed Jul. 28, 2022, the entire contents of which is hereby incorporated by reference.

FIELD OF THE DISCLOSURE

(2) The disclosure relates to a pinning device for the electrostatic pinning of a film on a chill roll, a corresponding casting unit as well as a stretching unit with such a casting unit.

BACKGROUND

(3) In the manufacture of thin plastics films, a film of a plastics melt is initially applied to a chill roll, on which it cools and solidifies. To ensure homogeneous cooling and thus homogeneous material properties, the film needs to be applied to the chill roll along its entire width uniformly, in fact continuously along its length.

(4) To avoid the film being deformed by mechanical pinning solutions, the generation of an electric field with high field strengths in proximity to the chill roll surface is known by means of electrodes, through said electric field the film is to pass through. Due to the electric field, the polar molecules of the film are aligned and the surface of the chill roll becomes electrically charged. An electrostatic attraction forms between the film and the chill roll so that the film applies itself to the chill roll uniformly and automatically.

(5) To generate such an electric field with strong electric field strengths, a very high voltage of 9 kV or more is required however at the electrodes that can thus result in sparkovers between the

electrodes and the chillroll. Such sparkovers damage the surface of the chill roll and also reduce the quality of the film produced.

SUMMARY

(6) There is provided a pinning device, a casting unit as well as a stretching unit in which sparkovers are reduced.

(7) For this purpose a pinning device for the electrostatic pinning of a film on a chill roll is provided, comprising a first coil unit, a second coil unit, a pinning region situated between the coil units, a high-voltage source as well as at least two electrodes, wherein at least the first electrode of at least two electrodes is strip-shaped. The electrodes run from the first coil unit to the second coil unit in the pinning region and high voltage is applied to both electrodes by the high-voltage source.

(8) By using two strip-shaped electrodes, to which high voltage is applied to both, a homogeneous electric field is generated that has particularly high electric field strengths, in particular maximum field strengths, at the point at which the film comes into contact with the chill roll. By focusing the electric field, the high voltage can be reduced in comparison to other solutions in order to generate the same electric field strength. By reducing the high voltage in the electrodes, the tendency for sparkovers is thus also reduced, thereby causing fewer sparkovers or avoiding sparkovers completely.

(9) For example, the high voltage has a voltage between 5 kV and 10 kV, in particular it is between 7 kV and 9 kV. The high voltage can even be up to 20 kV.

(10) In an embodiment, the second electrode of at least two electrodes is also strip-shaped or is wire-like, in particular a wire, thereby making it possible to improve the homogeneity of the electric field further.

(11) In an embodiment, a third electrode is provided, wherein the third electrode is strip-shaped or is wire-like, in particular a wire. By means of the third electrode, the homogeneity of the electric field is increased further.

(12) For example, the electrodes run parallel to each other, thereby making the electric field very homogeneous.

(13) Both electrodes can each comprise two lateral faces, wherein the electrodes each face each other with one of their lateral faces. In this way, the homogeneity of the electric field is increased further.

(14) The lateral faces run in particular parallel to each other.

(15) In one aspect, the electrodes have a spacing to each other of 3 mm to 15 mm, in particular 5 mm to 10 mm, in the pinning region, thereby achieving a particularly good focusing of the electric field.

(16) For example, the electrodes are the same width, thereby increasing the homogeneity of the electric field further. Differing widths are also conceivable, thereby making it possible to influence the position of the maximum field strengths.

(17) The width of the electrodes can be between 3 mm and 15 mm, in particular greater or equal to 4 mm and less than 13 mm, thereby achieving a very homogeneous electric field.

(18) To be capable of adapting the electrodes to the specifics of the casting unit, the electrodes can be offset in the direction of their widths or arranged to be flush.

(19) For example, the direction of the width equals the radial direction of the roller.

(20) In an embodiment, one, several or all of the electrodes are moveable in their longitudinal direction between the first coil unit and the second coil unit, in particular wherein at least one rotatable coil is provided in the first or the second coil unit, on which said rotatable coil one, several or all of the electrodes are partially coiled. As a result, deposits on said one, several or all electrodes that impair the quality of the electric field can be transported away.

(21) For example, a rotatable coil is located in each of the coil units for each of the electrodes, on said rotatable coil the corresponding electrode is coiled partially.

(22) In particular, at least one of the rotatable coils is powered.

- (23) In an embodiment, one, in particular the second electrode, or several of the electrodes are static, in particular wherein a mechanical tensioning device is provided in the second coil unit, wherein said one or several static electrodes are attached to the tensioning device in the second coil unit and fixed in position in the first coil unit. In this way, the complexity of the pinning device is simplified.
- (24) In an embodiment, a voltage application device is provided in at least one of the coil units, in particular in both coil units, said voltage application device connecting the electrodes electrically to said at least one high-voltage source, thereby ensuring a reliable application of high voltage to the electrodes.
- (25) For example, a joint high-voltage source is provided for both coil units or each coil unit has a separate high-voltage source.
- (26) In an embodiment, the voltage application device comprises a roller, on which the first electrode is guided, and/or the voltage application device has a curved guiding surface, along which the second electrode and/or the third electrode is guided. This facilitates a reliable and space-saving contacting of the electrodes.
- (27) For example, the external side of the guiding surface is a circular sector.
- (28) To prevent undesired vibrations of the electrodes, the second coil unit comprises a damping device that is configured to dampen the vibrations of the electrodes in the pinning region.
- (29) The electrodes, for example, are guided through the damping device out of the corresponding coil unit.
- (30) In particular, the vibrations to be dampened have an amplitude perpendicular to the lateral faces of the electrodes, in the direction of the width of the electrodes and/or are torsional vibrations about the longitudinal axis of the electrodes.
- (31) For example, the damping device comprises a damping roller whose rotational axis is perpendicular to the longitudinal direction and perpendicular to the direction of the width of the electrodes, wherein the damping roller is mounted movably in the direction of the width and rests on one edge of each of the electrodes, in particular on the edge facing away from the chill roll. Thus, a damping device can be realised with only a damping roller.
- (32) Alternatively or additionally, the damping device comprises two damping rollers whose rotational axes are perpendicular to the longitudinal direction and parallel to the direction of the width of the electrodes, wherein each of the electrodes rests on one of the damping rollers. In this way, each electrode is dampened separately, thereby ensuring that the electrodes do not mutually affect each other.
- (33) The damping rollers rest in particular on the lateral faces of the electrodes.
- (34) Both electrodes run through, for example, between both damping rollers.
- (35) In an embodiment, the pinning device comprises several insulating devices, in particular insulating conduits, insulating grommets and/or bushing insulators, which extend from the relevant coil unit towards the pinning region, wherein the electrodes run through the insulating devices, in particular wherein at least one of the insulating devices is provided for each of the electrodes on each of the coil units. By means of the insulating devices, sparkovers are reliably prevented outside the pinning region.
- (36) For example, the insulating devices are made of polyether ether ketone (PEEK), in particular made of annealed polyether ether ketone.
- (37) The insulating devices enclose the electrodes at least at the edge facing the chill roll.
- (38) In an embodiment, the insulating devices are attached in a vibration-damping manner to the damping device in order to minimise vibrations of the electrode guided therein.
- (39) In a further embodiment, low voltage is applied to at least one of the electrodes, in particular in addition to the application of the high voltage, thereby heating the electrode being applied with low voltage and thus reducing deposits and condensate on the electrode as such deposits and condensate primarily form on cold surfaces.

- (40) For example, due to the low voltage, an electric current can form between both coil units via each of the electrodes. In particular, the electrodes are not grounded so that the electric current comes to a standstill in the event of a crack in the electrode.
- (41) The low voltage has, for example, a maximum voltage of 150 V, for example between 60 V and 130 V, in particular 120 V. The electric current generated by the low voltage can have a strength of current that is between 1 A and 8 A.
- (42) For example, the pinning device comprises a low-voltage source for this. The high-voltage source and the low-voltage source can be designed as one device.
- (43) The object is also solved by means of a casting unit, containing at least a slot die, a chill roll and a pinning device as described previously, in particular wherein the chill roll is grounded.
- (44) Through the high voltage applied and the grounding of the chill roll, a small electric current is formed, termed pinning current.
- (45) In the case of a high voltage of 20 kV, the current is, for example, in the region of 25 mA and 30 mA. In the case of high voltage of 9 kV, a current can form ranging from about 9 mA to 12 mA.
- (46) If an electrode cracks, the high-voltage source is switched off. A crack in the electrode can be detected by edge detection.
- (47) The features and advantages described for the pinning device apply equally to the casting unit and vice versa.
- (48) The electrodes can have a spacing to the chill roll of at least the thickness of the film, for example at least 2 mm and/or a maximum of 20 mm.
- (49) Moreover, the object is solved by means of a stretching unit for producing a film, in particular a transverse direction orienter, a machine direction orienter and/or simultaneous stretching unit, comprising an oven and a casting unit as previously described.
- (50) The features and advantages described for the pinning device and/or the casting unit apply equally to the stretching unit and vice versa.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Additional features and advantages of the disclosure are found in the following description as well as the attached drawings to which reference is made. In the drawings:
- (2) FIG. 1 shows a stretching unit according to the disclosure comprising a casting unit according to the disclosure in a schematic view,
- (3) FIG. 2 shows the casting unit according to FIG. 1 comprising a pinning device according to the disclosure in a schematic, perspective view,
- (4) FIG. 3 shows schematically a cross section through the casting unit during operation,
- (5) FIG. 4 shows a perspective view of one of the coil units of the pinning device according to FIG. 2,
- (6) FIGS. 5,6 show schematically different parts of the pinning device within the second and first coil units as well as the electrode layout within the corresponding coil unit,
- (7) FIG. 7 shows a perspective view of the electrodes exiting the insulating devices of the coil unit according to FIG. 4,
- (8) FIG. 8 shows a perspective view of a damping device of the coil unit according to FIG. 4,
- (9) FIG. 9 shows a perspective section from the damping device of FIG. 8,
- (10) FIG. 10 shows a second embodiment of a damping device for a coil unit of a pinning device according to the disclosure,
- (11) FIG. 11 shows the damping device according to FIG. 10 in another view and partially opened, and
- (12) FIGS. 12, 13, 14 show schematic sections through a third, fourth and fifth embodiment of the

pinning device during operation, respectively.

DETAILED DESCRIPTION

(13) Lists having a plurality of alternatives connected by “and/or”, for example “A, B and/or C” are to be understood to disclose an arbitrary combination of the alternatives, i.e. the lists are to be read as “A and/or B and/or C” or as “at least one of A, B or C”. The same holds true for listings with more than three items.

(14) FIG. 1 shows a stretching unit **10** according to the disclosure for producing a film comprising a casting unit **12** and an oven **14**. The stretching unit **10** is for example a transverse direction orienter, a machine direction orienter and/or simultaneous stretching unit.

(15) To produce the film, the casting unit **12** in the present example initially generates a film **16** (also termed “cast film”) that is fed into the oven **14**. In the oven **14**, the film **16** is stretched in a known manner in the longitudinal direction and/or the transverse direction, thereby producing a mono-axially or bi-axially oriented film.

(16) In FIG. 2, the casting unit **12** is shown in an enlarged schematic view.

(17) The casting unit **12** shown here comprises a slot die **18**, a chill roll **20** as well as a pinning device **22**.

(18) The chill roll **20** is cooled.

(19) The slot die **18** is located above the chill roll **20** and is configured to continually apply a plastic melt onto the chill roll **20** which forms the film **16**. The plastic used is in particular a polyethylene terephthalate (PET) but is not limited to this.

(20) The film **16** is then applied by means of the pinning device **22** to the chill roll **20** uniformly. The chill roll **20** rotates here in the view according to FIG. 2 anticlockwise. The film **16** generated in this way is ultimately detached from the chill roll **20**, in the shown embodiment after about a three quarters rotation of the chill roll **20**, if necessary cooled further and fed into the oven **14**.

(21) The pinning device **22** is an electrostatic pinning device and comprises two electrodes, namely a first electrode **24** and a second electrode **26**, two coil units, namely a first coil unit **28** and a second coil unit **30**, a low-voltage source **31** as well as a high-voltage source **32**.

(22) In the shown embodiment, the low-voltage source **31** and the high-voltage source **32** are designed as one device. The low-voltage source **31** and the high-voltage source **32** can also be designed as separate devices.

(23) The coil units **28**, **30** are located in the axial direction of the chill roll **20** in front or behind the chill roll **20**. A pinning region **34** is configured between both coil units **28**, **30**, said pinning region **34** as seen in the axial direction of the chill roll **20** also includes the region of the film **16** on the chill roll **20**.

(24) The electrodes **24**, **26** are each both attached to each of the coil units **28**, **30** and extend between both coil units **28**, **30**, i.e. also through the pinning region **34**.

(25) In FIG. 3, a schematic view of the casting unit **12** is shown in the axial direction of the chill roll **20**. For simplification, only the electrodes **24**, **26** are shown from the pinning device **22**.

(26) The electrodes **24**, **26** are strip-shaped in the shown first embodiment, i.e. they have a width that is a lot longer than their thickness, in particular longer by one order of magnitude. In addition, they have a length in their longitudinal direction that is greater by one order of magnitude than their width.

(27) The electrodes **24**, **26** thus have two lateral faces, which extend in the longitudinal direction of the electrode **24**, **26** and in the direction of their width, as well as two edges. One of the edges is facing the chill roll **20** and one of the edges is facing away from the chill roll **20**. The direction between the edges of the same electrode **24**, **26** is termed the direction of the width of the electrodes **24**, **26**.

(28) In the first embodiment, both electrodes **24**, **26** have the same width, for example between 3 mm and 15 mm, in particular a width that is greater or equal to 4 mm and less than 13 mm. For example, a width of 12.7 mm is conceivable.

(29) In the shown embodiment, the electrodes **24, 26** each point to the other electrode **24, 26** with one of its lateral faces. The lateral faces and therefore also both electrodes **24, 26** run parallel to each other, at least in the pinning region **34**.

(30) In the pinning region **34**, the electrodes **24, 26** have a spacing to each other of 3 mm to 15 mm, in particular 5 mm to 10 mm.

(31) Both electrodes **24, 26** are arranged to be flush to each other in the direction of their width, i.e. a connecting line of their edges is perpendicular to the lateral faces.

(32) In particular, the direction of the width for one of the electrodes **24, 26** equals the radial direction of the chill roll **20**.

(33) The electrodes **24, 26** are located in the rotational direction of the chill roll **20** after the slot die **18**. For example, the electrodes **24, 26** have a spacing to the chill roll **20** of at least the thickness of the film **16**, for example at least 2 mm. At the most, they have a spacing to the chill roll **20** of 20 mm.

(34) High voltage is applied to the electrodes **24, 26** by the high-voltage source **32**. The high voltage is, for example, between 5 kV and 10 kV, in particular between 7 kV and 9 kV.

(35) The chill roll **20** is grounded so that a strong electric field forms in the pinning region **34**, through said electric field the film **16** is guided.

(36) In addition, a weak electric current forms which is called a pinning current. In the case of a high voltage of 20 kV, the current is, for example, at about 30 mA. In the case of high voltage of 9 kV, a current of about 12 mA can form.

(37) If one of the electrodes **24, 26** cracks, the high-voltage source is switched off. A crack in one of the electrodes **24, 26** can be detected by the high-voltage source by means of edge detection.

(38) Similarly, low voltage is applied to at least one, in particular both electrodes **24, 26** by the low-voltage source **31**. The low voltage is, for example, less than 150 V, in particular between 60 V and 130 V, for example 120 V. The high voltage and the low voltage are in cascade accordingly.

(39) In FIG. 4, the second coil unit **30** is shown opened. The second coil unit **30** is located in relation to FIG. 2 on the left side of the chill roll **20**.

(40) The first coil unit **28** is constructed identically on the right side of the chill roll **20**, merely as a mirror image. Any differences that exist will be indicated separately.

(41) The second coil unit **30** comprises a housing **36**, a rotatable coil **38**, a voltage application device **40**, a first deflecting roller **42**, a mechanical tensioning device **44**, a second deflecting roller **46**, a damping device **48** as well as two insulating devices **50**.

(42) In FIG. 5, the coil unit **30** is shown simplified even further; only the two electrodes **24, 26** as well as the components for guiding the electrodes **24, 26** are shown. For example, the housing **36**, the damping device **48** and the insulating devices **50** are not shown.

(43) The first electrode **24** is coiled partially on the rotatable coil **38** and runs from the rotatable coil **38** to the voltage application device **40**.

(44) The voltage application device **40** is connected to the high-voltage source **32** electrically and comprises a roller **52** as well as a curved guiding surface **54**.

(45) The roller **52** guides the first electrode **24** and in doing so high voltage provided by the high-voltage source **32** is applied to the first electrode **24**.

(46) Then, the first electrode **24** runs to the first deflecting roller **42** and from there out of the coil unit **30** towards the pinning region **34**.

(47) The second electrode **26** is attached to the tensioning device **44** in the second coil unit **30**. The tensioning device **44** comprises for this a tension bolt **56** that can be actuated by means of a motor **58** (FIG. 4).

(48) The second electrode **26** runs then to the voltage application device **40**. On the voltage application device **40**, the second electrode **26** runs along the guiding surface **54**.

(49) The guiding surface **54** has a curved external side for guiding the second electrode **26**. For example, the curved external side is a circular sector.

(50) High voltage is applied to the second electrode **26** via the contact with the guiding surface **54**, said high voltage being provided by the high-voltage source **32**.

(51) Then, the second electrode **26** runs via the second deflecting roller **46**, through which it is arranged parallel to the first electrode **24**. Both electrodes **24**, **26** then run out of the second coil unit **30** through the pinning region **34** and finally into the first coil unit **28**, which is reproduced in FIG. **6** in a reduced form.

(52) The first electrode **24** runs in the reverse order as described for the second coil unit **30**, initially via the first deflecting roller **24**, the roller **52** of the voltage application device **40** and then onto the rotatable coil **38**.

(53) The voltage application device **40** of the first coil unit **28** is also connected to the high-voltage source **32** electrically. It is however also conceivable that each of the coil units **28**, **30** have a separate high-voltage source.

(54) Both rotatable coils **38** of the coil units **28**, **30** can be powered. In this way, the first electrode **24** can be moved between the first coil unit **28** and the second coil unit **30**. For example, the first electrode **24** is uncoiled from the rotatable coil **38** of the first coil unit **28**, runs through the pinning region **34** into the second coil unit **30**, where it is coiled by the rotatable coil **38** of the second coil unit **30**.

(55) In addition, the tensile force or the mechanical tension of the first electrode **24** can be set by means of the powered rotatable coil **38**.

(56) In contrast, the second electrode **26** is static in the shown embodiment. It runs into the first coil unit **28** and is guided by the second deflection roller **46** to the voltage application device **40**. Then, the second electrode **26** is fixed in position by means of the end piece **60** in the first coil unit **28**. Thus, in contrast to the second coil unit **30**, the first coil unit **28** does not comprise a tensioning device **44**.

(57) The tensile force or the mechanical tension of the second electrode **26** is set by means of the tensioning device **44**.

(58) It is however conceivable that rotatable coils are also provided for the second electrode **26** instead of the end piece **60** and the tensioning device **44**, similar to the rotatable coils **38** for the first electrode **24**. In this way, the second electrode **26** is also then configured to be moveable.

(59) For example, a separate rotatable coil can be provided for each electrode **24**, **26** in each coil unit **28**, **30**.

(60) It is also conceivable that several of the electrodes **24**, **26** are coiled on the same rotatable coil **38**. In this case, a joint rotatable coil **38** is provided, for example, which comprises different regions for the different electrodes **24**, **26**. The mechanical tensions of the different electrodes **24**, **26** can then be set by a tensioning device for each of the electrodes **24**, **26**.

(61) As shown in FIG. **4**, the electrodes **24**, **26** run out of the coil units **28**, **30** through the damping devices **48** and ultimately out of the coil units **28**, **30** through the insulating devices **50**.

(62) The insulating devices **50** thus extend from the damping device **48** to the pinning region **34**, as can be recognised in FIG. **2**.

(63) In particular, the insulating devices **50** extend along that region of the chill roll **20**, on which no film **16** is applied.

(64) In FIG. **7**, a perspective view of the end of the insulating devices **50** is shown that points to the pinning region **34**. In the shown representation, the electrodes **24**, **26** exit from the insulating devices **50** and then run in the pinning region **34**.

(65) The insulating devices **50** are for example insulating conduits, insulating grommets and/or bushing insulators. They have an elongated cross section that is almost rectangular. For example, the corners are rounded or the short sides are formed by curved sections.

(66) The inner diameters of the insulating devices **50** in their length are thus the width of the electrodes **24**, **26** so that each one of the electrodes **24**, **26** can be guided in one of the insulating devices **50**.

(67) In the shown embodiment, the insulating devices **50** surround the received electrodes **24**, **26** along their circumference each completely.

(68) It is however conceivable that the electrodes **24**, **26** are not completely enclosed as long as they are enclosed from each other and/or towards the chill roll by the respective insulating device **50**.

(69) The insulating devices **50** are made of a plastic, in particular polyether ether ketone, PEEK. In particular, the PEEK is an annealed PEEK. Insulating devices **50** made of annealed PEEK have proven to be particularly durable.

(70) The insulating devices **50** are attached at the exit of the damping device **48** by means of an attachment device **74**, as can be seen in FIG. **8**.

(71) The attachment device **74** has, for example, vibration-damping elements **76**, such as dampers made of an elastic material, between which the insulating devices **50** are clamped. The clamp can be set, for example, using a threaded fastener.

(72) The damping devices **48** are provided with a side arm **62** that extends from the housing **36** of the corresponding coil unit **28**, **30** in the direction towards the pinning region **34**.

(73) In particular, the length of the side arm **62** is adjustable, for example by means of a motor and a rack-and-pinion gear.

(74) The damping device **48** is shown enlarged in FIGS. **8** and **9**. The damping device **48** comprises a support frame **64** and a damping roller **66**.

(75) The support frame **64** has two supports **68** that are located on different sides of the two electrodes **24**, **26**. In other words, both electrodes **24**, **26** run through between the two supports **68**.

(76) The damping roller **66** is located above, i.e. on the side of the electrodes **24**, **26** facing away from the chill roll **20**. The damping roller **66** is mounted rotatably on the support frame **64** about a rotational axis **R1**, wherein the rotational axis **R1** is perpendicular to the longitudinal direction of the electrodes **24**, **26** as well as perpendicular to the direction of the width of the electrodes **24**, **26**.

(77) The damping roller **66** rests on the upper edges of both electrodes **24**, **26**, i.e. the edges facing away from the chill roll **20**. Thus, it rotates with the movement of the first electrode **24** when being coiled or uncoiled.

(78) In the direction of the width of the electrodes **24**, **26**, the damping roller **66** is mounted moveably, for example spring-loaded.

(79) During the operation of the stretching unit **10**, an electric high-voltage field is generated between the electrodes **24**, **26** and the chill roll **20** by means of the pinning device **22**, in particular the two parallel, strip-like electrodes **24**, **26** to which high voltage has been applied. The film **16** passes through the electric field due to the rotary motion of the chill roll **20**.

(80) The surface of the chill roll **20** is electrically charged by the electric field and the polar molecules of the plastics material of the film **16** align. As a result, an electrostatic attraction forms between the chill roll **20** and the film **16**. Through this force of attraction, the film **16** is applied to the chill roll **20** uniformly.

(81) During operation, the first electrode **24** (and if applicable with the provision of corresponding rotatable coils **38** also the second electrode **26**) is moved continuously along its longitudinal direction, i.e. unwound in the first coil unit **28** and wound in the second coil unit **30**, or vice versa. In this way, deposits caused by the evaporating plastics material of the film **16** are transported away, thereby increasing the homogeneity of the electric field.

(82) Deposits are also reduced by the low voltage applied to the electrodes **24**, **26**. The low voltages forms a current, termed a heater current, through each of the electrodes **24**, **26** from the first coil unit **28** to the second coil unit **30** (or vice versa) with a strength of current between 1 A and 8 A.

(83) For example, the strength of current through an electrode **24**, **26** formed as a 3 mm strip is between 2 and 2.5 A and through an electrode **24**, **26** formed as a 12.7 mm strip is approximately 7 A. In using a wire as the electrode **26**, the currents are lower.

(84) The electrodes **24, 26** are not grounded so that a crack in an electrode **24, 26** interrupts the current.

(85) The electrodes **24, 26** heat up as a result of the current, thereby causing the evaporation of the plastics material of the film **16** to condense less strongly as condensate on the electrodes **24, 26**.

(86) In spite of the mechanical tension applied to the electrodes **24, 26** by the rotatable coil **38** and the tensioning device **44**, the electrodes **24, 26** can vibrate, for example with an amplitude perpendicular to the lateral faces, in the direction of the width of the electrodes **24, 26** and/or about the longitudinal axis of the electrodes **24, 26**. The damping device **48** dampens or prevents these possible vibrations of the electrodes **24, 26** so that the electric field remains constant. The quality of the pinning is thus increased further.

(87) It is also conceivable that vibrations are prevented by an undulating layout of the electrodes **24, 26** in the pinning region **34**. For this purpose, the pinning device **22** comprises then a bracket with pins about which the electrodes **24, 26** are guided.

(88) By using both strip-shaped electrodes **24, 26**, the electric field is at a maximum on the film surface at the point which first comes into contact with the chill roll **20**. In this way, the high voltage can be reduced without lowering the quality of the pinning, sparkovers are reduced as a result.

(89) In FIGS. **10** and **11**, a second embodiment of the casting unit **12** according to the disclosure and/or pinning device **22** according to the disclosure are shown. This corresponds substantially to the first embodiment so that only the differences are discussed hereinafter. Identical and functionally equivalent parts are provided with the same reference signs.

(90) The FIGS. **10** and **11** show a damping device **48** of a second embodiment of the pinning device **22**.

(91) In this second embodiment, the damping device **48** comprises two damping rollers **66** that each are attached to a corresponding bearing block **70**.

(92) Each bearing block **70** is spring-mounted perpendicular to the lateral faces of the electrodes **24, 26** so that the damping rollers **66** are also spring-mounted.

(93) Each of the damping rollers **66** is rotatable about a rotational axis **R2** that is perpendicular to the longitudinal direction and parallel to the direction of the width of the electrodes **24, 26**. In other words, the rotational axis **R2** extends parallel to the lateral faces, but perpendicular to the longitudinal direction of the electrodes **24, 26**.

(94) Both electrodes **24, 26** run through between both damping rollers **66**, wherein each one of the damping rollers **66** contacts the lateral face of one of the electrodes **24, 26**. In particular, both electrodes **24, 26** still do not contact each other.

(95) In other words, one of the electrodes **24** or **26** is allocated to each damping roller **66**, with said electrode **24** or **26** running on its lateral face.

(96) It is also conceivable that in addition to both damping rollers **66** of the second embodiment, a damping roller **66** is also present as shown for the first embodiment in order to increase the damping effect further.

(97) In FIGS. **12** to **14**, further embodiments of the pinning device **22** are shown which substantially correspond to the first embodiment. Therefore, only the differences are discussed hereinafter and the same parts and parts with the same function are given the same reference signs. In addition, they can also have the features of the second embodiment.

(98) In FIG. **12**, a third embodiment of the pinning device **22** is shown in a section similar to that of FIG. **3**.

(99) In this embodiment, the strip-shaped electrodes **24, 26** do not have the same width, but have different widths.

(100) In the shown embodiment, the first electrode **24** has a smaller width than the second electrode **26**. Similarly, it is conceivable that the second electrode **26** has a smaller width than the first electrode **24**.

- (101) Moreover, the second electrode **26** is offset in relation to the first electrode **24** in the direction of its width. In this way, the electric field can be adapted to the specifics of the pinning device **22** or the casting unit **12**, for example the slot die **18**.
- (102) It is also conceivable that the pinning device **22** has three or more strip-shaped electrodes that run parallelly in the pinning region **34**. One, several or all of these electrodes can be moveable in their longitudinal direction.
- (103) In FIG. **13**, a fourth embodiment of the pinning device **22** is shown in a section similar to that of FIG. **3**.
- (104) In this fourth embodiment, the second electrode **26** is not strip-shaped, but is wire-like, in particular formed as a wire. Almost any other cross-section of the second electrode **26** is also conceivable.
- (105) The second electrode **26** runs in the fourth embodiment at the same spacing to the chill roll **20** as the first electrode **24**, more specifically the edge of the first electrode **24** facing the chill roll **20**.
- (106) It is however conceivable that the second electrode **26** is arranged further away from the chill roll **20**.
- (107) In FIG. **14**, a fifth embodiment of the pinning device **22** is shown in a section similar to that of FIG. **3**.
- (108) In this fifth embodiment, the pinning device **22** comprises a third electrode **72** in addition to the first electrode **24** and the second electrode **26**. The pinning device **22** of the fifth embodiment thus has three electrodes **24**, **26**, **72**.
- (109) The first electrode **24** is strip-shaped. The second electrode **26** and the third electrode **72** are formed to be wire-like, in particular as wires. Almost any other cross-section of the second electrode **26** and/or the third electrode **72** is also conceivable.
- (110) The second and the third electrodes **26**, **72** are arranged on different sides of the first electrode **24**. The second and the third electrodes **26**, **72** can have the same spacing to the first electrode **24**.
- (111) In the shown example, however, the first electrode **24** is not situated directly between the second and third electrodes **26**, **72**, i.e. the first electrode **24** does not form an imaginary connecting line between the second electrode **26** and the third electrode **72**.
- (112) The first electrode **24** is offset towards the chill roll **20** in relation to the second and third electrodes **26**, **72**.
- (113) The second and the third electrodes **26**, **72** can have the same spacing to the chill roll **20**.
- (114) Through the different arrangements and constructions of the electrodes **24**, **26**, **72**, the electric field can be suitably formed for the respective application, in particular for the respective plastics material being used.
- (115) It is also conceivable that the second electrode **26** and/or the third electrode **72** are strip-shaped.
- (116) Similarly, high voltage and/or low voltage can be applied to each of the three electrodes **24**, **26**, **72**, which can be designed either to be moveable in the longitudinal direction or static.

Claims

1. A pinning device for an electrostatic pinning of a film on a chill roll, comprising: a first coil unit, a second coil unit, a pinning region situated between the first and second coil units, a high-voltage source as well as at least two electrodes, wherein at least a first electrode of the at least two electrodes is strip-shaped, wherein the at least two electrodes run from the first coil unit to the second coil unit in the pinning region, and wherein high voltage is applied to the at least two electrodes by the high-voltage source.
2. The pinning device according to claim 1, wherein a second electrode of the at least two

electrodes is strip-shaped or is wire-like.

3. The pinning device according to claim 1, wherein a third electrode is provided, wherein the third electrode is strip-shaped or is wire-like.

4. The pinning device according to claim 1, wherein the at least two electrodes run parallel to each other or each comprise two lateral faces, wherein the at least two electrodes each face each other with one of their lateral faces.

5. The pinning device according to claim 1, wherein the at least two electrodes have a spacing to each other of 3 mm to 15 mm in the pinning region.

6. The pinning device according to claim 1, wherein at least one of the at least two electrodes have a same width, the at least two electrodes have different widths or the width of the at least two electrodes is between 3 mm and 15 mm.

7. The pinning device according to claim 1, wherein the at least two electrodes are arranged offset in a direction of their width or arranged to be flush.

8. The pinning device according to claim 1, wherein one, several or all of the at least two electrodes are moveable between the first coil unit and the second coil unit in their longitudinal direction.

9. The pinning device according to claim 8, wherein at least one rotatable coil is provided in the first and in the second coil unit, on said at least one rotatable coil, wherein several or all of the at least two electrodes are partially coiled.

10. The pinning device according to claim 1, wherein a voltage application device is provided in at least one of the first and second coil units, said voltage application device connecting the at least two electrodes electrically to said at least one high-voltage source.

11. The pinning device according to claim 10, wherein the voltage application device comprises at least one of a roller, on which the first electrode is guided, or a curved guiding surface, along which at least one of a second electrode or a third electrode is guided.

12. The pinning device according to claim 1, wherein at least one of the first coil unit or the second coil unit comprises a damping device that is configured to dampen vibrations of the at least two electrodes in the pinning region.

13. The pinning device according to claim 12, wherein the damping device comprises a damping roller whose rotational axis is perpendicular to a longitudinal direction and perpendicular to a direction of the width of the at least two electrodes, wherein the damping roller is mounted movably in the direction of the width and rests on one edge of each of the at least two electrodes.

14. The pinning device according to claim 13, wherein the damping device comprises two damping rollers whose rotational axis are perpendicular to the longitudinal direction and parallel to the direction of the width of the at least two electrodes, wherein each of the at least two electrodes touches one of the two damping rollers.

15. The pinning device according to claim 1, wherein the pinning device further comprising several insulating devices, which extend from the corresponding first and second coil units towards the pinning region, wherein the at least two electrodes run through the insulating devices.

16. The pinning device according to claim 15, wherein at least one of the several insulating devices is provided for each of the at least two electrodes on each of the first and second coil units.

17. The pinning device according to claim 15, wherein the insulating devices are attached in a vibration-damping manner to the damping device.

18. The pinning device according to claim 1, wherein low voltage is applied to at least one of the at least two electrodes.

19. A casting unit comprising a slot die, a chill roll and a pinning device, the pinning device comprising a first coil unit, a second coil unit, a pinning region situated between the first and second coil units, a high-voltage source as well as at least two electrodes, wherein at least a first electrode of the at least two electrodes is strip-shaped, wherein the at least two electrodes run from the first coil unit to the second coil unit in the pinning region, and wherein high voltage is applied to the at least two electrodes by the high-voltage source.

20. A stretching unit for producing a film, comprising an oven and the casting unit according to claim 19.
